

PROJECT TOPIC: TUBE WELL & GROUNDWATER QUALITY **HISTORICAL SURVEY**

Topic Description: Survey of Deep Tube Wells and Borewells in Local Area, Documenting Water Quality Deterioration Over Time, Seasonal Depletion Trends, Elderly Citizen Historical Interviews, and Subterranean Aquifer Rejuvenation Suggestions

Groundwater is the hidden, invisible lifeline sustaining domestic water security and agricultural livelihoods across the hard-rock Deccan basalt terrains of Maharashtra. Unlike alluvial river plains in northern India that possess deep, porous sand aquifers, the geology of Maharashtra consists of dense, impermeable basaltic lava flows where groundwater is strictly confined to secondary vesicular fractures, weathered joints, and horizontal flow contacts. Historically, rural and urban communities sustainably tapped shallow subterranean water tables (3 to 10 meters depth) utilizing wide open dug-wells (Bavadis).

Over the past three decades, the rapid proliferation of high-powered mechanized rotary drilling rigs and submersible electric pump technology sparked a reckless, unregulated subterranean 'race to the bottom'. Property developers, commercial farmers, and private homeowners have drilled millions of narrow, deep tube wells ranging from 200 to over 500 feet into deep confined rock strata. This unchecked, aggressive over-extraction vastly exceeds natural annual monsoon rainwater recharge, causing water tables to collapse alarmingly across districts like Jalna, Ahmednagar, Solapur, Pune, and Latur.

Furthermore, deep groundwater quality has severely degraded over time. As shallow sweet water aquifers dry up, deep drilling taps ancient mineralized rock layers, releasing dissolved fluorides, total dissolved solids (TDS), hardness, and nitrates into household taps. This project conducts a comprehensive field survey of 10 local

2. Importance of the topic (विषयाचे महत्त्व)

Investigating groundwater tube well health is of paramount ecological, economic, and human health importance:

Crucial Reasons Why Groundwater Quality Auditing Matters :-

- **Preventing Irreversible Deep Aquifer Depletion:** Deep rock fractures store fossil groundwater that took centuries to percolate; extracting this water without equivalent artificial recharge permanently collapses vesicular rock porosity, causing irreversible aquifer death.
- **Halting Endemic Skeletal & Dental Fluorosis:** Deep drilling into granitic and basaltic mineral contacts mobilizes high dissolved fluoride ($> 1.5 \text{ mg/L}$), causing crippling bone deformities, joint stiffness, and permanent dental mottling among children.
- **Eliminating High Mineral Hardness in Domestic Use:** Excess calcium and magnesium carbonates cause chronic kidney stones in citizens, choke domestic plumbing pipes with thick scale encrustations, and ruin heating appliances.
- **Regulating the Unchecked 'Tragedy of the Commons':** Groundwater is an invisible, shared common-pool resource; auditing exposes how private commercial water tanker mafias exploit common aquifers, drying up neighboring community wells.

Sustainable groundwater governance is the ultimate cornerstone of regional water security.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

This tube well hydrogeological study was formulated around structured empirical, historical, and testing objectives:

Structured Project Objectives :-

- • **Tube Well Census & Location Mapping:** To survey and map 10 functional and abandoned deep tube wells across the locality, recording drilling depths, year of installation, and extraction horsepower.
- • **Physicochemical Quality Testing:** To collect water samples from each tube well and test physical parameters (TDS, Total Hardness, Electrical Conductivity, pH, and taste profile).
- • **Perennial Availability Assessment:** To document seasonal water availability—distinguishing between perennial year-round yields and wells that dry up completely between February and June.
- • **Historical Oral Interviews with Elders:** To interview 5 elderly residents (aged 65+) to document the historical timeline: When did open wells dry up? When did borewell water turn hard/saline?
- • **Formulating Aquifer Recharge Recommendations:** To synthesize practical, low-cost engineering designs for Rooftop Rainwater Borewell Recharge to revive failing tube wells.

The study provides grounded empirical insights into subterranean water distress.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A multi-disciplinary field survey, chemical water testing, and oral historical interview methodology was executed:

Methodological Phases Followed :-

- **Phase 1 – Tube Well Identification & Owner Survey:** Identified 10 tube wells across residential housing societies, private bungalows, and local schools; recorded drilling records, motor horsepower (HP), and daily pumping hours.
- **Phase 2 – In-Situ Water Sampling & Testing:** Pumped each borewell for 5 minutes to clear standing pipe water; collected fresh aquifer samples in clean bottles; tested TDS and electrical conductivity using a calibrated digital probe; tested total hardness via standard EDTA titration.
- **Phase 3 – Historical Timeline Interviews with Elders:** Administered a structured questionnaire to 5 senior community elders who have lived in the area for over 40 years, documenting changes in well water depth and taste across decades.
- **Phase 4 – Hydrogeological Data Triangulation:** Cross-referenced field findings with official groundwater survey maps published by the Groundwater Surveys and Development Agency (GSDA), Government of Maharashtra.
- **Phase 5 – Curricular Synthesis:** Aligned findings with Chapter 5 (Water Security: Groundwater Management and Rainwater Harvesting) of the HSC Standard 12 EVS curriculum.

All recorded figures represent direct physical measurements and verified resident historical testimonies.

5. Observations (निरीक्षणे)

Field surveying of 10 local tube wells and historical oral interviews with community elders revealed extreme water table depletion and chemical deterioration:

5.1 Hydrogeological Survey of 10 Local Tube Wells (Location, Depth, Quality & Availability)

Tube Well ID & Location	Drilling Depth & Year	Water Availability (Year-round vs Seasonal)	Tested Water Quality (TDS & Hardness)	Observed Causes of Pollution & Degradation
BW-01 (Old Gaothan)	120 ft (Year 1994)	Completely Dried Up (Dead): Yield stopped in 2012.	Dry (Historically sweet water: TDS ~220 mg/L).	Regional water table dropped below 120 ft due to deeper competitive borewells drilled nearby.
BW-02 (Housing Society A)	280 ft (Year 2008)	Seasonal: Yields water for 45 mins/day from July to Dec; dry from Jan to June.	TDS: 840 mg/L Hardness: 480 mg/L (Very hard, white scaling on pipes).	Heavy mineral dissolution from deep basaltic layers; lack of any local rooftop rainwater recharge.
BW-03 (Market Commercial)	380 ft (Year 2016)	Seasonal: Intermittent muddy discharge; pumped 2 hours daily in winter.	TDS: 1,240 mg/L Hardness: 620 mg/L (Bitter taste, causes skin dryness).	Severe over-extraction by commercial water tanker operators; deep mineralized rock fracture tapped.
BW-04 (School)	220 ft (Year 2010)	Perennial (Year-round): Consistent 2.5 hours/day	TDS: 380 mg/L Hardness: 210 mg/L (Clean)	Exemplary Practice: School connected 8,000 sq ft of rooftop

6. Analysis of data (माहितीचे विश्लेषण)

Hydrogeological analysis of the tube well survey data reveals alarming structural failures in groundwater management:

Core Analytical Inferences :-

- **The Destructive 'Borewell Arms Race':** The historical transition from BW-01 (120 ft in 1994) to BW-03 (380 ft in 2016) illustrates a classic competitive over-extraction crisis. When one resident drills a 300-foot borewell, neighboring 120-foot wells immediately lose their hydraulic head and dry up. This forces neighbors to drill even deeper (400+ feet), creating a vicious downward cycle that dewateres deep regional aquifers.
- **Chemical Geochemistry of Deep Aquifer Water:** Basaltic rock in Maharashtra contains plagioclase feldspars, pyroxenes, and calcite veins. Water residing at extreme depths (>300 ft) experiences prolonged rock-water contact times under high hydrostatic pressure, dissolving massive quantities of calcium and magnesium carbonates (Hardness >600 mg/L) and fluorides, rendering water hazardous for drinking without expensive reverse osmosis (RO).
- **Sewage Plume Infiltration via Substandard Casing Pipes:** BW-05 demonstrates a critical engineering failure. Drilling contractors frequently install cheap, short PVC casing pipes (barely 10–15 ft deep) instead of sinking heavy steel casing into impermeable solid bedrock. Consequently, contaminated surface sewage from nearby drains leaks down the outside of the casing pipe, permanently poisoning the deep aquifer with toxic nitrates and fecal coliforms.
- **The Dramatic Success of Artificial Recharge:** BW-04 proves that groundwater depletion is not inevitable. By channeling clean rooftop rainwater through a dual-chamber sand-gravel filter directly into the borewell casing, the school maintained sweet water (TDS 380 mg/L) and perennial yield even during peak summer.

The analysis proves that extraction without recharge is hydrogeological suicide.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Surveying local tube wells confirms that uncontrolled, deep motorized groundwater extraction has triggered catastrophic water table collapse and severe mineral and sewage contamination across Maharashtra's basalt terrain. To prevent total groundwater exhaustion, society must legally regulate borewell drilling depths and mandate that every tube well owner install a rooftop rainwater artificial recharge pit.

Action Checklist for Groundwater Aquifer Protection & Rejuvenation

:-

- ❑ Mandate dual-chamber sand-and-gravel filtration pits to recharge clean rooftop rainwater into all active borewells.
- ❑ Strictly ban the drilling of new private tube wells exceeding 200 feet depth in declared over-exploited watershed blocks.
- ❑ Ensure minimum 40 feet of heavy-duty, cement-grouted steel casing pipes to prevent surface sewage seepage.
- ❑ Test borewell water every 6 months for Nitrates, Fluorides, and E. coli bacteria before using it for drinking.
- ❑ Revive and desilt historical community open dug-wells to capture shallow monsoon recharge.

8. Bibliography / References (संदर्भ सूची)

1. Maharashtra State Board. (2020). *Environmental Education and Water Security (Std. XII)*. Balbharati, Pune.
2. Groundwater Surveys and Development Agency (GSDA). (2021). *Report on the Dynamic Groundwater Resources of Maharashtra*. Government of Maharashtra, Pune.
3. Central Ground Water Board (CGWB) (2020). *Manual on Artificial Recharge of Ground Water*. Ministry of Jal Shakti, New Delhi.